Commentary on Tasks 1- 4 in Coursework 1.2

This analysis uses the USArrests dataset in R which consists of arrests per 100,000 residents for murder, assault and rape in each of the 50 states in 1973. This analysis examines the quality of the data by checking for duplicates and missing values, the descriptive statistics of the data, and how it is distributed in this dataset. We need to check the relationships between the variables via visual plots and the limitations of the dataset while interpreting the data.

In task 1, we were required to check for the validity and quality of values in the dataset.

#validate_data(USArrests)

This function was created to test if there were duplicates, non-numeric analysis columns, negative values, and values outside plausible ranges. This function returned FALSE results for these factors. However, passing this function's checks does not guarantee accuracy, as this validation is limited to the conditions in which the function was programmed. Therefore, it is not suitable to guarantee that the dataset is error-free.

In task 2, Using,

calculate_stats <- function(column) {

c(

Min = min(column),

Max= max(column),

Mean= mean(column),

Median = median(column),

SD= sd(column),

IQR= IQR(column)

)

}

We calculated descriptive statistics to examine the centre, spread, and possible skewness of the four variables. Assault had the highest variability, with an interquartile range of 140 and a standard deviation of 83.34; this indicates the differences in assault arrest rates between states. The means of Murder, Assault, and Rape were higher than their medians, which indicates a possible Right skewness. This distribution was examined further with a visualization choice: histogram.

Premasundari and Yamini (2019) pointed out that the results obtained from the USArrests dataset are helpful in predicting the probability of high crime rates in various states in the US through visualization of the crime analysis.

Histograms and boxplots were plotted because they provide more visual information about the distributions. As initially suspected, Murder and Rape appeared right-skewed on the plot, Assault appeared to be bimodal with two higher frequencies, UrbanPop looks symmetric, with the highest frequency in the 60-70% interval. The boxplots showed variation in Assault arrest rates and showed two potential outliers for rape representing Alaska (44.5) and Nevada (46.0). These instances may have contributed to the right-skewness of the Rape distribution, but it is too excessive to say they are the determinant for errors.

Frequency labels were included in the histogram charts, and the quartile values were included as labels on the boxplots for easier reading. A limitation is that the patterns on the histogram depend on the selected bin widths, and potential outliers on boxplots do not automatically determine data error. The visualizations generally help to examine patterns and differences between the 50 states, but the results should be analyzed in view of the dataset’s limitations.

We calculated and plotted Pearson and Spearman correlation coefficients to check for the relationships between Murder, Assault, Rape, and UrbanPop. J.C.F De Winter et al. (2016) say that the Pearson correlation coefficient is used to measure the linear

relationship between two continuous random variables and is used when the data follows a normal distribution, especially when the correlation is thought to be strong and Spearman correlation measures monotonic relationship between two continuous random variables and is used when the data severely do not follow a normal distribution. Correlation heatmaps were used to compare the coefficients, while ggpairs() combined the scatterplots, correlation values, and distributions for more visual results.

These two methods showed Murder and Assault to have the strongest relationship (Pearson = 0.802; Spearman = 0.817), while Murder and UrbanPop had the weakest relationship (Pearson = 0.070; Spearman = 0.107). The “Murder-Rape relationship” was stronger using Spearman (0.679) than Pearson (0.564); this suggests that the relationship may not be completely linear. But this contrast alone should not define the exact shape of the relationship.

Both correlation methods give us more knowledge of the variables’ relationship, but none of them proves that one crime happens because of the other. Also, arrest rates may be determined by how frequently crimes are reported and how the police handle them with possible bias, so they may not reflect the true rates of crime. Lastly, since this data is from 1973, the results may not completely apply to the current year, so we should be careful when using these data to predict or make decisions about crimes or use a police force resource.

The USArrests dataset has a lot of limitations that should be considered when analyzing its results. The data was recorded in 1973, and it may not show accurate crime patterns in the current years. We also need to consider the difference in policing across the decades. The dataset also has records of arrest rates and not the number of crimes committed. This suggests that the differences between states may be flawed due to crime reporting rates, police enforcement, and certain cultural and racial biases rather than actual differences in crime committed.

This dataset only contains four variables and does not consider other social factors like education, security, economic inequality, and demographic differences that may influence the rate of crimes and arrests. The correlations also cannot explain why these

differences happen. The correlation also does not explain if one incident was caused by another, so the positive relationship between Murder and Assault does not mean that one is directly responsible for the other.

These results may help keep trends of patterns for future investigation, but they should not be considered alone to guide crime prevention or prediction policies or be used alone as a police resource. There should be more recent data, additional variables, and further research per state, including the rate of crimes committed to arrests made in order to support accurate investigation and decisions.

Conclusively, the analysis showed differences in arrest rates across 50 states in the US. Assault showed the highest variability. Murder and UrbanPop had the lowest positive relationship, while Murder and Assault had the highest positive relationship. The dataset's year, the limited variables, and dependence on just arrest rates and not crimes committed inclusive should be analyzed carefully.

References

De Winter, J.C.F., Gosling, S. D., & Potter, J. (2016). "COMPARING THE PEARSON AND SPEARMAN CORRELATION COEFFICIENTS ACROSS DISTRIBUTIONS AND SAMPLE SIZES: A TUTORIAL USING SIMULATIONS AND EMPIRICAL DATA." *Psychological Methods*, 21(3), 273-290. Available at https://doi.org/10.1037/met0000079 [https://arxiv.org/pdf/2408.15979]. [Accessed October 23, 2026]

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